



UNIVERSITY OF SCIENCE - VNUHCM

FACULTY OF INFORMATION TECHNOLOGY

SOFTWARE TESTING

HW4 - GUI & USABILITY TESTING

Software Testing Project Report

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July 22, 2025

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1 Group Information

Group ID: 07

Member Name	Student ID	Assigned Tables	Status
Cao Uyển Nhi	22127310	Registration Page	Done
Lưu Thanh Thuý	22127410	Checkout Page	Done
Nguyễn Phước Minh Trí	22127424	Categories Page	Done
Võ Lê Việt Tú	22127435	Home Page	Done
Trần Thị Cát Tường	22127444	Categories Management Page	Done

2 GUI Testing with AI Tools

2.1 Tool Name

The AI tool used for this task is **ChatGPT (GPT-4)**, accessed via the web interface at chat.openai.com. This generative AI model is capable of natural language understanding and visual analysis, making it suitable for exploratory GUI evaluation when combined with interface screenshots and contextual prompts.

2.2 Prompts Used

To simulate AI-based GUI testing, I submitted interface screenshots of the “Categories Management” page from the Toolshop web application into ChatGPT. I also provided the full GUI checklist in Excel format to support structured evaluation.

The AI tool was instructed to assess the interface both visually and against each line item in the checklist, ensuring full coverage of UI consistency, alignment, and usability issues.

The following screenshots were used as visual input:

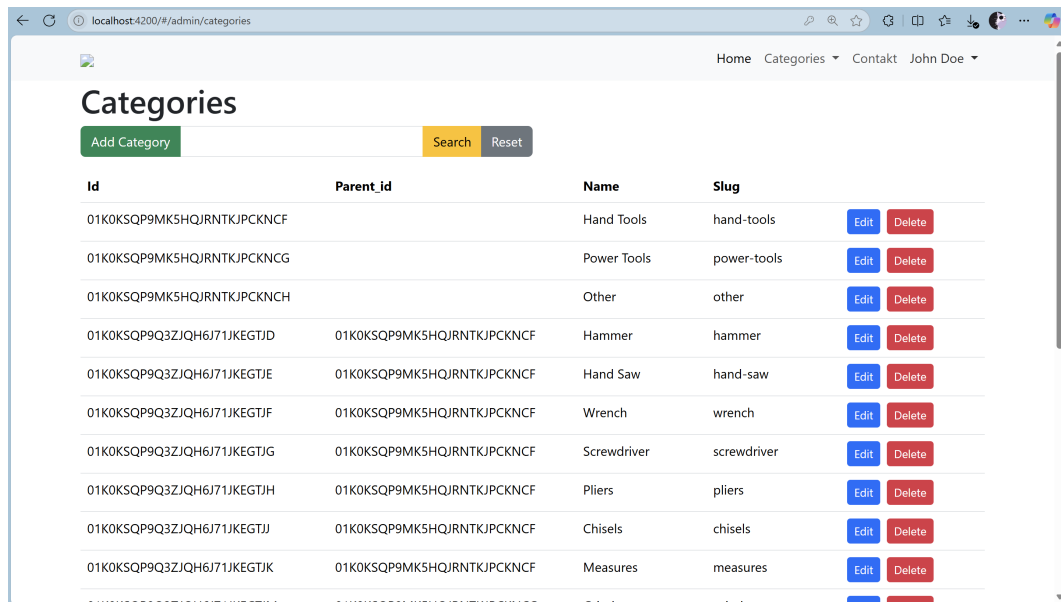


Figure 1: Categories Management Page - 1

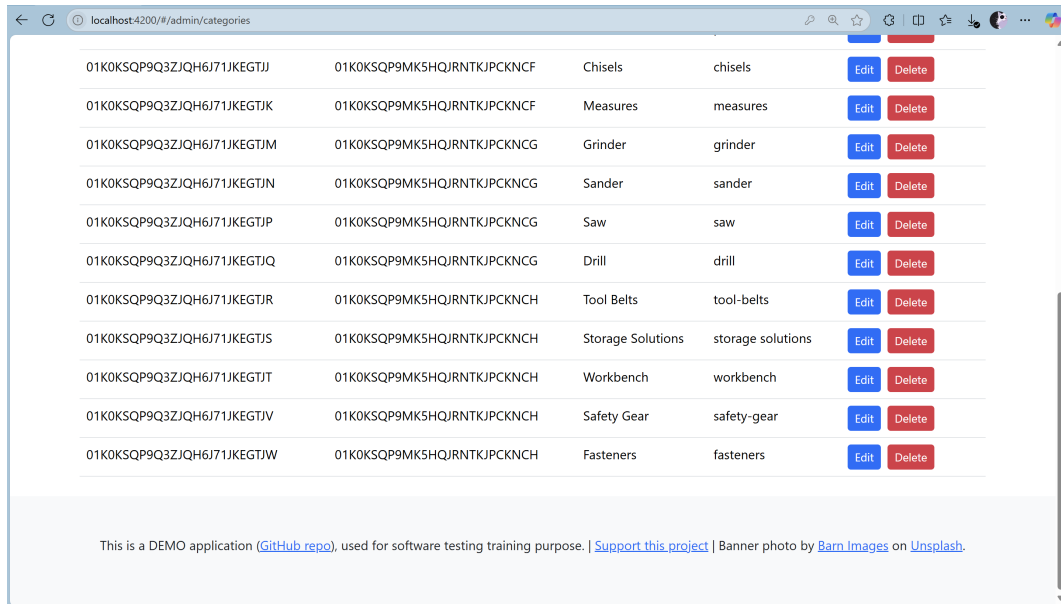


Figure 2: Categories Management Page - 2

The following is the used prompt:

“plain text You are a senior software tester. I want you to evaluate a GUI based on both a checklist and provided interface screenshots.

- The GUI checklist contains columns: “No.”, “Checkpoint”, “Yes”, “No”, “Remarks”.
- The interface is from a web application used to manage product categories.
- Please review each checklist item in the context of the screenshots and give an assessment in the format: Yes/No + a short justification in ‘Remarks’.
- Focus your evaluation on the following aspects:
 - Navigation and Link Functionality
 - Page Structure and Layout
 - Color Consistency
 - Font Usage and Typography
 - Button and UI Element Behavior
 - Search Bar Placement and Functionality
 - Sorting and Table Data Presentation
 - Spelling and Language Accuracy
 - Alignment and Spacing
 - Visual Grouping of Components
 - Header/Footer Display
 - Hyperlink Implementation (Websites and Emails)

Your task: Go through the checklist item by item, and for each one, provide: → A “Yes” or “No” judgment

→ A brief explanation in the Remarks column

Refer to the screenshots for visual reference.

2.3 Validation and Refinement of AI-Generated Results

Each AI-generated response was carefully reviewed to ensure accuracy, completeness, and alignment with the actual GUI of the system under test. The validation process was performed manually using the provided screenshots of the “Categories Management” screen and followed a structured and iterative approach:

- **Accuracy Check** For every checklist item, I compared the Yes/No judgment and the accompanying remarks generated by the AI with what was visibly present in the interface. This included verifying whether certain links existed, checking color consistency, identifying misalignments, and confirming whether fonts or UI components were used consistently. If the AI’s assessment did not align with what was clearly visible, I flagged it for revision.
- **Prompt Refinement** In cases where the AI response was vague, too general, or descriptive without drawing a clear conclusion, I refined the prompt. I explicitly asked for binary answers (Yes or No) and required the AI to include a justification in the Remarks column. For instance, instead of accepting a vague statement like “the buttons are mostly consistent,” I adjusted the prompt to require: “Are all buttons consistently styled? Answer Yes/No and explain briefly why.”
- **Context Reinforcement** Some checklist items depended on contextual understanding (e.g., determining whether the user knows their current page location). In these cases, I clarified the question by highlighting specific UI regions (e.g., the header or navigation bar) or by giving examples of what elements might serve as breadcrumbs or page indicators. This helped the AI to focus its attention on relevant parts of the screenshot and avoid misinterpretation.
- **Consistency Review** After generating responses for multiple checklist items, I reviewed the entire output to ensure the tone, terminology, and level of detail were consistent throughout the evaluation. I ensured that similar issues received consistent treatment, and that the explanations remained concise but informative. This was important for integrating the AI feedback into a formal test report.

This validation and refinement process significantly improved the reliability of AI-generated feedback. It allowed the AI to perform more like a human tester—systematic, precise, and aligned with real-world usability expectations. Ultimately, it ensured that the final output could be trusted and used for informed decision-making in GUI quality assessment.

2.4 Test Case Sources: Manual vs. AI-Generated

The test cases used in this evaluation originated from two sources, with the majority of the testing effort grounded in manual design and execution, and AI serving as a complementary verification tool.

2.4.1 Manually Created Test Cases

The primary basis for GUI evaluation was a comprehensive checklist developed manually using the file Web GUI Checklist template.xlsx. This checklist contains over

30 structured criteria, systematically covering various GUI testing dimensions, including:

The primary basis for GUI evaluation was a comprehensive checklist developed manually using the file *Web GUI Checklist template.xlsx*. This checklist contains over 30 structured criteria, systematically covering the following testing areas:

- Link navigation and behavior (Verifying whether hyperlinks direct to the intended pages, checking for orphan or dead-end pages – 1.1.1 → 1.1.4, 2.1.1)
- Text alignment and layout spacing (Ensuring content and headers are properly aligned, no overlaps or layout breaks – 1.3.5, 1.3.6)
- Typography and font consistency (Checking font type, size, and emphasis across buttons, headers, and body text – 1.3.1 → 1.3.3)
- Button styling and interactive behavior (Validating button color consistency, size, and placement for actions like Edit/Delete – 1.2.7, 2.2.5)
- Search bar placement and usability (Verifying visibility, location, and whether it stands out appropriately – 1.3.8, 2.1.3)
- Hyperlink styling and functionality (Confirming that web/email links are clickable and styled correctly on hover/visit – 1.1.5, 1.2.5, 1.2.6)
- Sorting and presentation of table data (Checking whether data appears sorted or provides any sorting functionality – 2.2.6)
- Logical grouping of UI components (ensuring related UI elements such as action buttons are grouped logically – 2.2.5)

These items follow standard GUI and usability testing practices and were executed carefully by analyzing actual interface screenshots.

2.4.2 AI-Generated Test Cases

While the checklist formed the core of the testing process, ChatGPT was employed to validate and reinforce the findings. Using the same screenshots and checklist content, the AI was prompted to independently assess each item. In doing so, the AI helped identify additional perspectives or overlooked edge cases, particularly in areas such as:

- 1.2.4: Detecting whether bold, italic, or hyperlink text had clearly distinguishable colors compared to regular content.
- 1.3.3: Evaluating subtle inconsistencies in font usage and visual hierarchy between headings and body text.
- 1.3.6: Checking for precise text alignment across labels, titles, and content blocks.
- 1.3.8: Judging the visual prominence and accessibility of the search bar.
- 2.2.6: Assessing whether the data presented in tables appeared to be sorted correctly or lacked sorting functionality.

These AI-based observations did not replace manual testing but rather served to confirm results and enhance confidence in the final assessment. They also provided a heuristic layer of feedback that complemented the structured checklist approach.

3 Self-Assessment

Criteria	Outcomes	Grade	Self-Assessed
1	Checklist of one GUI	30	30
2	AI tools	20	20
3	User survey and feedback	30	30
3.1	Questions (10)	10	10
3.2	Feedback (7)	10	10
3.3	Report	10	10
4	BrowserStack	30	30
5	Bug report	30	30
5.1	GUI	10	10
5.2	Usability	10	10
5.3	Cross-platform	10	10
	Total	140	140